EdgeER

System Design and Engineering Evaluation of a Low-Cost Edge-Assisted Multipatient Reassessment Prototype

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Abstract

EdgeER is an investigational edge-assisted multipatient monitoring and reassessment prototype that integrates a Raspberry Pi edge node, shared continuous photoplethysmography (PPG) processing, a FastAPI service, SQLite persistence, registry-backed device assignment, encounter management, packet audit logging, and a browser dashboard. The evaluation combines deterministic software validation with exploratory bench measurements. All 83 automated tests passed; the packet-validation matrix passed 45 of 45 cases with zero false acceptances or false rejections; device lifecycle checks passed 31 of 31 cases; three complete workflow scenarios passed; database persistence and integrity checks passed; ten simulated reconnect cycles recovered successfully; and 20 shared-PPG architecture assertions passed. Across twelve 60-second in-process load runs at 1, 5, 10, and 20 simulated devices, 16,205 of 16,217 submitted packets were accepted, with all non-accepted packets arising from deliberate fault injection. In ten paired observations against a Zacurate 500 Series consumer pulse oximeter, EdgeER showed mean absolute differences of 1.30 bpm for heart rate and 1.20 percentage points for SpO2. Three controlled PPG-derived respiratory-rate trials demonstrated confidence-gated availability, including withholding of an ambiguous fast-breathing estimate. Logged supply measurements corresponded to approximately 1.97-2.15 W at idle and 1.97-2.82 W while active. These findings establish a reproducible engineering baseline but do not constitute clinical validation or medical-device performance evidence.

Keywords: edge computing, PPG, FastAPI, SQLite, signal-quality gating, multipatient workflow, consumer-reference comparison, power characterization

Intended use. EdgeER is an engineering prototype for demonstrating embedded sensing, distributed communication, quality-aware data handling, and reassessment workflow software. It is not a medical device, is not clinically validated, and is not intended for patient-care decisions.

1. Introduction

EdgeER addresses the engineering problem of coordinating repeated measurements, device state, and reassessment workflow across multiple active encounters. A physical or simulated edge node submits timestamped measurements and quality metadata to a central service. The server validates the packet, enforces device-to-encounter assignment, records accepted and rejected traffic, persists encounter state, and exposes a dashboard for priority management, recheck reminders, action logging, offline awareness, and structured handoff generation.

The evaluation emphasizes software correctness, resilience, persistence, scalability, and reproducibility, supplemented by a small consumer-reference comparison, controlled respiratory-rate trials, and logged input-power observations. The bench evidence is exploratory: it uses a consumer pulse oximeter, manual respiratory counts, and a limited number of observations rather than calibrated clinical reference equipment or a participant study.

2. System Design

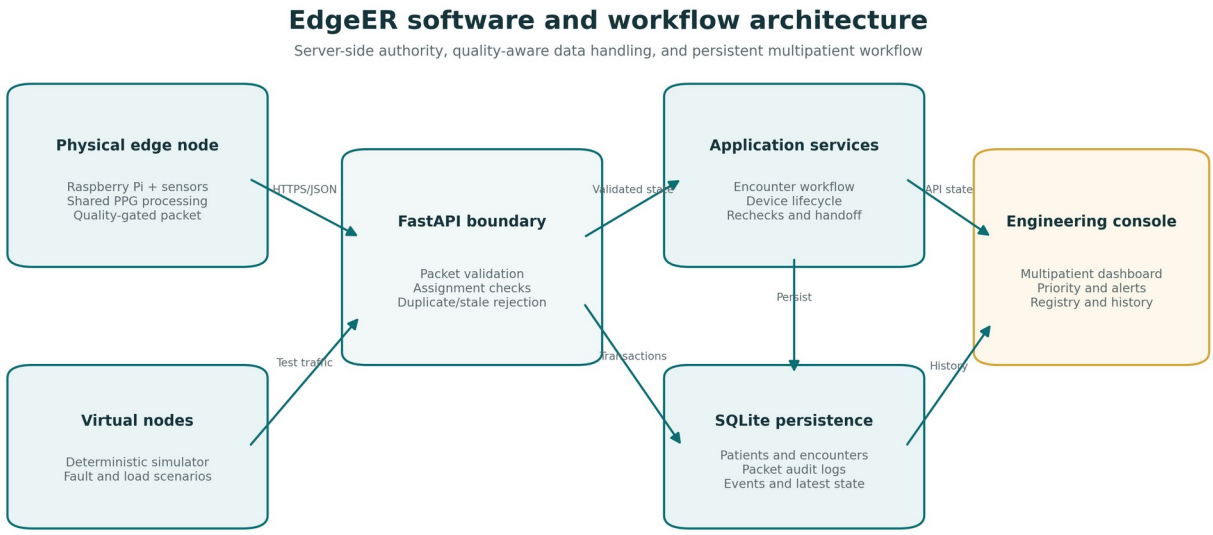


Figure 1. EdgeER software and workflow architecture.

Server-side authority governs device assignment, encounter state, packet acceptance, workflow priority, and historical persistence. Accepted packets update the current encounter and latest-state records; rejected packets remain auditable with explicit rejection reasons.

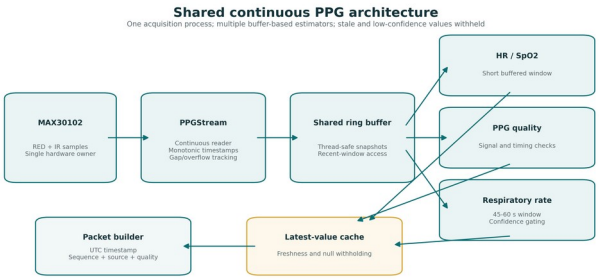


Figure 2. One-reader, multiple-consumer PPG pipeline.

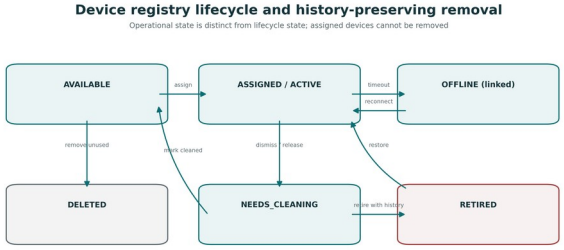


Figure 3. History-preserving device lifecycle.

The edge node uses one continuous MAX30102 acquisition owner and a thread-safe timestamped ring buffer. Heart-rate/SpO2 processing, PPG quality analysis, and experimental respiratory-rate estimation consume independent snapshots. Cached measurements are withheld when unavailable, stale, or insufficiently supported. Device lifecycle is distinct from operational connectivity: offline, cleaning, and sensor-fault states do not imply retirement, while retirement preserves historical references and blocks future assignment and packets until explicit restoration.

3. Evaluation Methodology

The software-validation suite used deterministic seeds, isolated temporary SQLite databases, UTC timestamps, monotonic duration measurements, explicit process cleanup, and separate accounting for expected fault injections and unexpected failures. Raspberry Pi hardware and the normal development database were not accessed by the automated Windows suite. Bench observations were evaluated separately from the software tests and were not used to alter validation thresholds.

Environment item	Recorded value
Operating system	Windows 10 (build 26200)
Python	3.11.9
Processor	AMD64; 16 logical CPUs
Memory	31.1 GiB
FastAPI / Starlette	0.116.1 / 0.47.3
SQLite	3.45.1
Deterministic seed	20260720
Load protocol	1, 5, 10, 20 devices; 3 x 60 s each

The software evidence covered automated tests, packet validation, device lifecycle, complete encounter workflows, database persistence and integrity, simulator scalability, API latency, simulated offline/reconnection behavior, dashboard contracts and screenshots, and shared-PPG software assertions. The bench evidence comprised ten paired heart-rate and SpO2 observations against a consumer fingertip pulse oximeter, three previously recorded 60-second respiratory-rate trials with manual counts, and logged voltage/current ranges used to calculate input power.

4. Functional Correctness Results

Validation area	Result
Automated tests	83 / 83 passed
Packet validation	45 / 45; 0 false accepts; 0 false rejects
Device lifecycle	31 / 31 checks passed
Workflow scenarios	3 / 3 complete scenarios passed
Database persistence	6 / 6 checks; integrity ok; 0 FK violations
Reconnection	10 / 10 cycles passed
Dashboard contracts	22 / 22 checks; 10 screenshots
Shared PPG	20 / 20 assertions passed

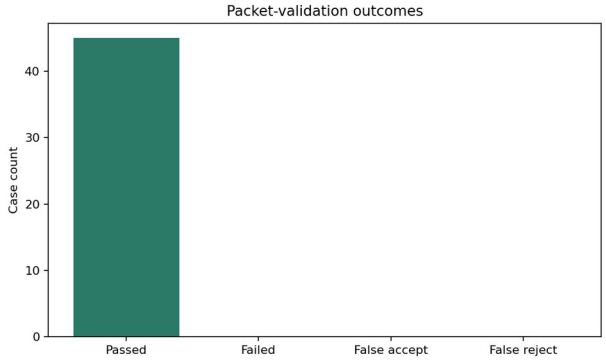


Figure 4. Packet-validation outcomes.

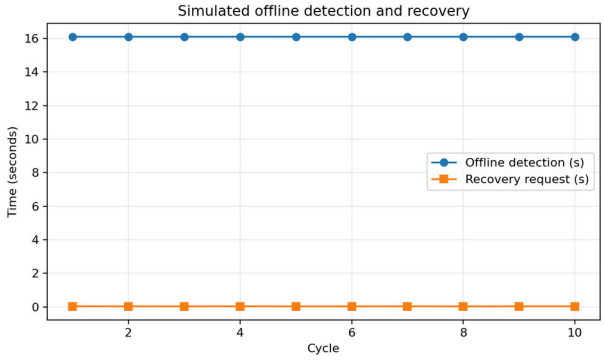


Figure 5. Simulated offline detection and recovery.

The packet matrix included missing identifiers, invalid types and ranges, stale and future timestamps, duplicate and out-of-order sequences, unknown and retired devices, wrong assignments, inactive encounters, valid quality-gated RR packets, optional-field handling, and deliberately malformed payloads. Every expected acceptance and rejection was observed, and every rejection contained a clear reason.

Scenario	Checks	Lifecycle verification	Result
Workflow scenario 1	15 / 15	Old encounter preserved; new encounter created	PASS
Workflow scenario 2	15 / 15	Old encounter preserved; new encounter created	PASS
Workflow scenario 3	15 / 15	Old encounter preserved; new encounter created	PASS

The workflow scenarios exercised registration, assignment, packet ingestion, recheck state, priority transitions, timeline actions, offline/reconnect state, dismissal, handoff generation, device release, readmission, reassignment, and restart restoration. Persistence checks confirmed active and closed encounters, priority state, recheck state, retired-device state, and legacy lifecycle migration. SQLite integrity_check returned ok and foreign_key_check reported no violations.

5. Scalability and Resource Results

Devices	Runs	Attempted	Accepted	Mean ms	p95 ms	Req/s	CPU	Peak MiB	DB +MiB
1	3	3,797	3,794	48.5	235.7	21.05	1.93%	83.0	3.00
5	3	3,844	3,841	55.0	254.8	21.33	1.94%	113.3	4.38
10	3	4,470	4,467	40.5	54.1	24.78	2.26%	147.6	5.93
20	3	4,106	4,103	46.8	248.5	22.78	2.05%	179.1	5.52

Across twelve runs, 16,217 packets were submitted and 16,205 accepted. The twelve non-accepted packets were intentional fault injections; no unexpected rejection or request error occurred. Mean latency remained between 40.5 and 55.0 ms, throughput between 21.05 and 24.78 requests/s, average CPU between 1.93% and 2.26%, and peak memory between 83.0 and 179.1 MiB.

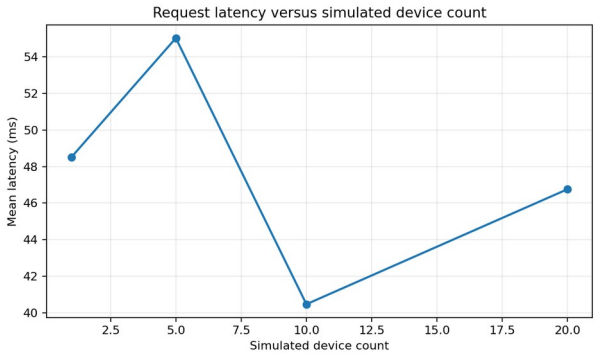


Figure 6. Mean request latency by simulated device count.

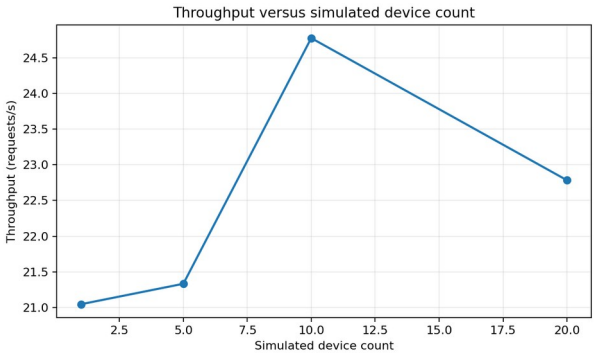


Figure 7. Throughput by simulated device count.

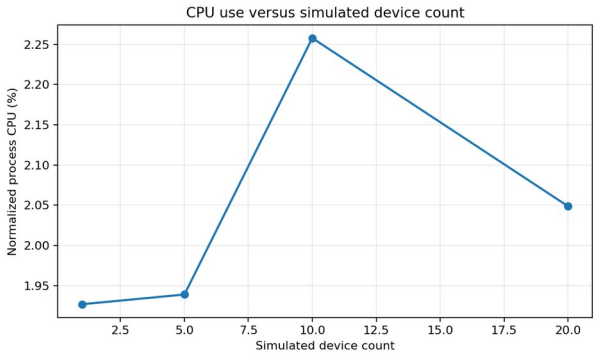


Figure 8. Average CPU use by simulated device count.

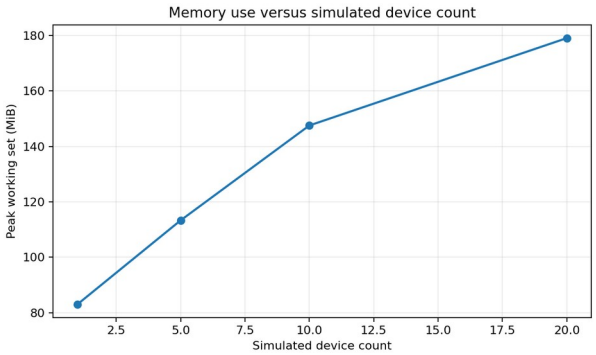


Figure 9. Peak memory use by simulated device count.

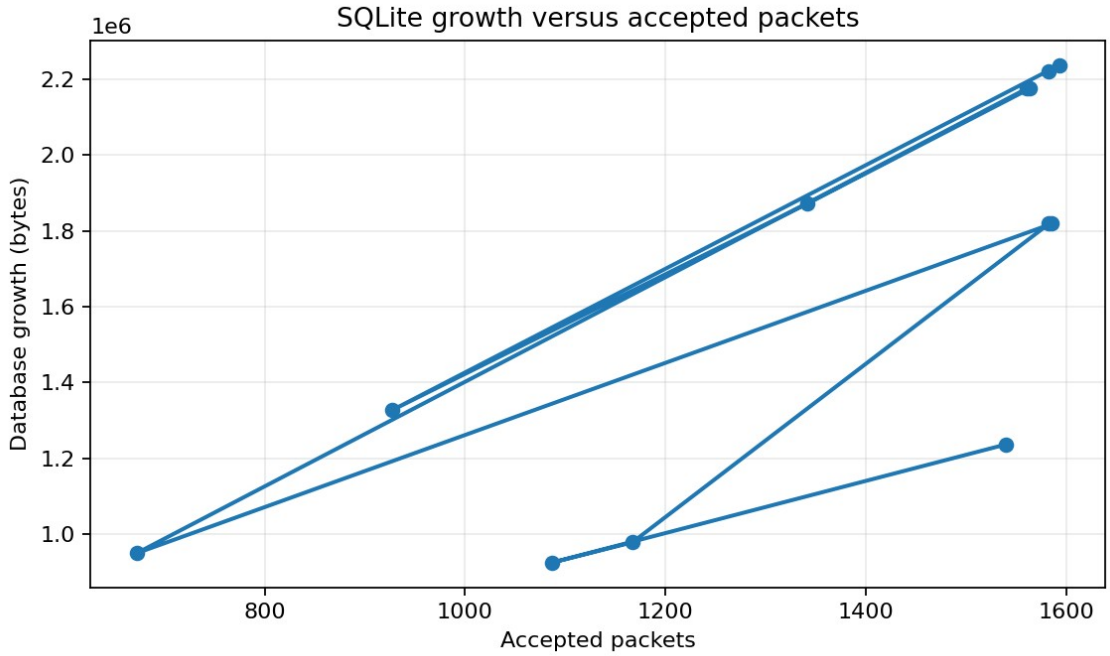


Figure 10. SQLite growth as accepted-packet volume increased.

Interpretation boundary. These are in-process FastAPI TestClient measurements on one Windows workstation. They characterize software regression and resource behavior, not physical-network capacity or deployment-scale throughput.

6. API Performance and Reconnection

Method	Endpoint	Mean ms	p95 ms	p99 ms
GET	/api/devices	2.29	2.73	2.87
GET	/api/patients	1.08	1.19	1.27
GET	/api/history	0.86	1.01	1.07
GET	/api/packet_logs?limit=50	2.33	2.75	2.85
GET	/api/encounters/{id}	1.98	2.33	2.45
GET	/api/handoff/{id}	2.16	2.47	2.58
POST	/api/vitals	37.43	40.34	43.28
POST	/api/action	14.96	16.44	17.34

Reconnection metric	Result
Cycles	10 / 10 passed
Configured offline timeout	15 s
Mean observed detection	16.106 s
Mean recovery	36.90 ms
p95 recovery	38.67 ms

Reconnection metric	Result
Assignment preserved	10 / 10
Duplicate rejected	10 / 10
Stale packet rejected	10 / 10

The simulated reconnection sequence stopped packet transmission, observed the offline transition, resumed the same assigned node, and verified the first accepted packet after recovery. Assignment remained intact, stale queued packets and duplicates were rejected, and no new active encounter was created.

7. Dashboard and Interface Evidence

Dashboard validation combined 22 reproducible API/template contract checks with ten browser screenshots across 1366 x 768, 1920 x 1080, tablet, and narrow-laptop viewports. The capture reported no browser console warnings or errors. The evidence verified registry-backed assignment, explicit no-device states, separate RR and patient-status presentation, priority/current-encounter separation, device lifecycle display, handoff availability, and consistent table structure.

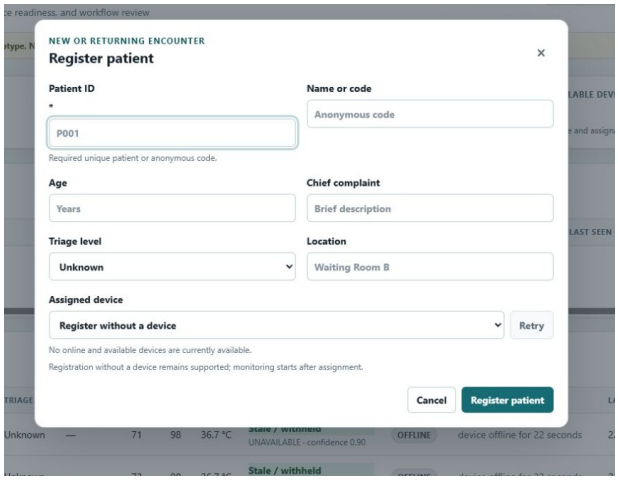


Figure 11. Registry-backed patient registration.

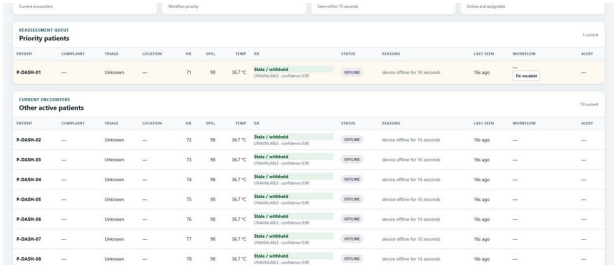


Figure 12. Device registry and assignment state.

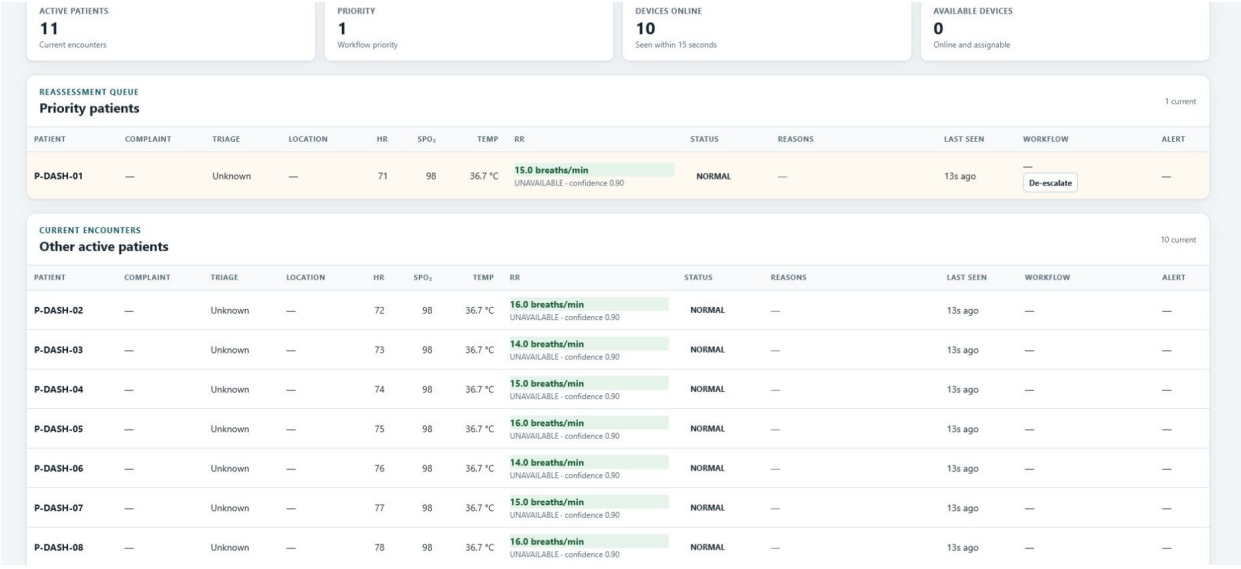


Figure 13. Multipatient dashboard under a ten-device simulated workload.

8. Exploratory Bench Evaluation

8.1 Consumer pulse-oximeter comparison

Ten paired observations were collected over a 20-minute bench session using EdgeER and a Zacurate 500 Series Fingertip Pulse Oximeter as a consumer reference device [1]. Conditions included seated rest, seated recovery, light walking, and one minor-motion observation. Readings were paired at the displayed-value level rather than from synchronized waveforms. The comparison therefore characterizes informal agreement under the recorded conditions and is not a clinical accuracy study.

Metric	Heart rate	SpO2
Paired observations	10	10
Mean absolute difference	1.30 bpm	1.20 percentage points
Root mean square difference	1.45 bpm	1.41 percentage points
Mean signed difference (EdgeER - reference)	-0.10 bpm	-0.40 percentage points
Maximum absolute difference	2 bpm	2 percentage points
Within +/-2 units	10/10 (100%)	10/10 (100%)

Heart-rate readings tracked the consumer reference closely across the observed range of 68-88 bpm. SpO2 values differed by no more than two percentage points in any trial; the narrow 95-99% range limits interpretation of correlation and does not establish performance outside normal bench conditions. The full paired observations are preserved with the report data.

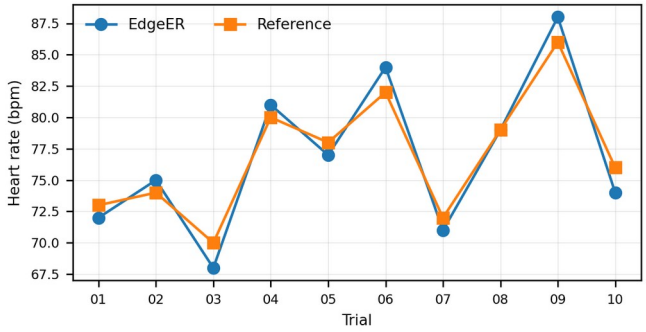


Figure 14. Paired heart-rate observations.

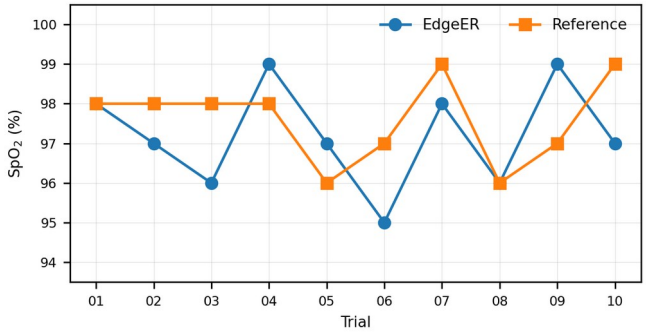


Figure 15. Paired SpO2 observations.

8.2 PPG-derived respiratory-rate trials

Three previously recorded 60-second trials compared the PPG-derived respiratory-rate candidate with a manual breath count. The current estimator transmitted only usable GOOD or MODERATE results. The fast-breathing candidate was numerically close to the manual count but was withheld because competing respiratory-band peaks reduced confidence. This behavior illustrates that the quality gate represents signal ambiguity rather than numerical closeness to an unavailable ground truth at runtime.

Condition	Manual RR	PPG candidate	Confidence	Quality	Packet value	Absolute difference
Normal	12.0	10.6	0.910	GOOD	10.6 sent	1.4
Slow	7.0	7.0	0.851	MODERATE	7.0 sent	0.0
Fast	22.0	21.624	0.211	LOW CONF.	withheld (null)	0.376*

*The fast-trial difference refers to the internal diagnostic candidate; the transmitted respiratory-rate value was null. For the two eligible trials, the mean absolute difference was 0.70 breaths/min and both were within +/-2 breaths/min. The three-trial sample is descriptive and too small for population-level performance claims.

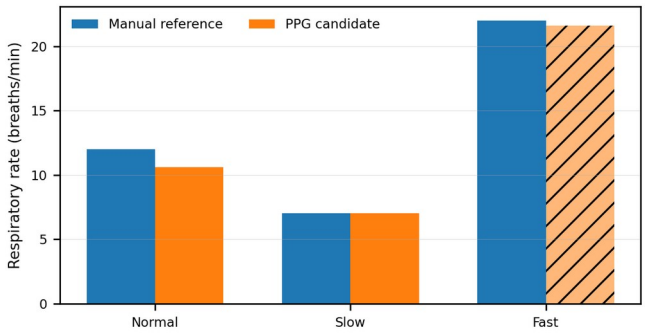


Figure 16. Manual counts and diagnostic PPG candidates; hatching marks the withheld trial.

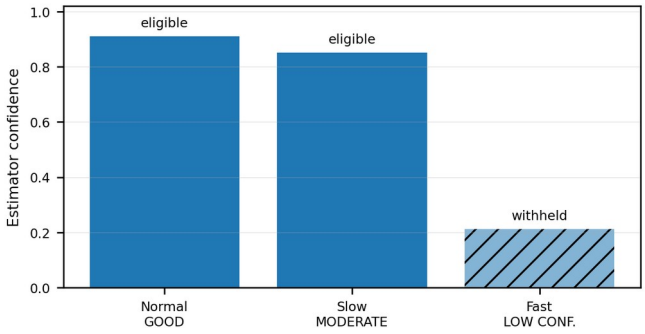


Figure 17. Estimator confidence and packet eligibility.

8.3 Logged input-power characterization

Logged supply voltage remained between 5.365 and 5.373 V in both operating states, a span of approximately 0.15% around the midpoint. Input power was calculated from the logged voltage and current values using P = VI. Idle current ranged from 0.368 to 0.401 A and typically remained near 0.368-0.372 A. Active current ranged from 0.368 to 0.525 A and most frequently appeared near 0.414, 0.418, and 0.444 A.

State	Voltage range	Current range	Calculated power range	Typical calculated power
Idle	5.365-5.373 V	0.368-0.401 A	1.97-2.15 W	1.98-2.00 W
Active	5.365-5.373 V	0.368-0.525 A	1.97-2.82 W	2.22, 2.24, 2.38 W

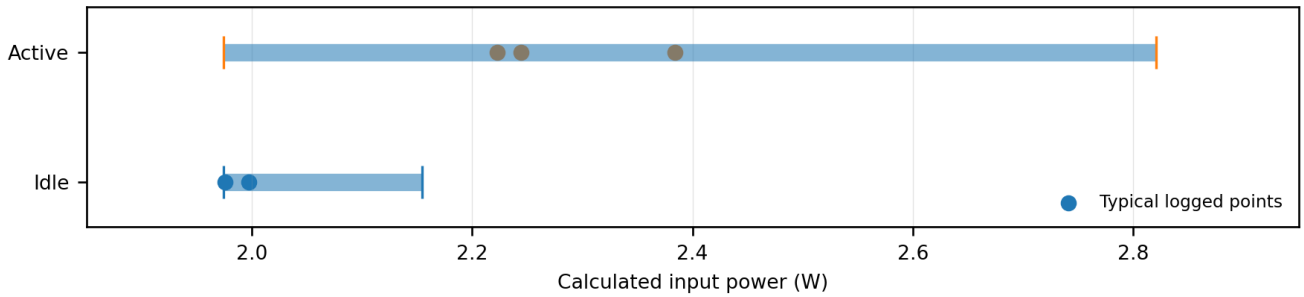


Figure 18. Calculated input-power ranges and typical logged operating points.

The active upper bound reflects a short observed current peak; the listed typical points better represent the recurring active readings in the log. These measurements characterize the tested node and supply arrangement rather than a complete energy-consumption or battery-runtime model.

9. Limitations and Conclusion

Limitation	Interpretation
In-process transport	TestClient excludes real Wi-Fi, router, and network-stack effects.
Short load duration	Each scalability run lasted 60 seconds; long-duration drift was not measured.
Single workstation	Performance reflects one Windows workstation and its local scheduler.
Synthetic scenarios	Simulator data demonstrate workflow behavior, not physiological realism.
Consumer reference	Ten displayed-value pairs from one bench session do not constitute calibrated or clinical validation.
Respiratory-rate sample size	Three controlled trials describe gating behavior but do not support population-level accuracy estimates.
Power evidence	Logged ranges characterize one node and supply arrangement; no calibrated energy or battery-runtime study was performed.
No clinical claims	The results do not establish diagnostic accuracy, clinical safety, or patient-care effectiveness.

The evidence demonstrates a coherent engineering implementation: one shared PPG acquisition owner with buffer-based consumers, deterministic packet enforcement, history-preserving device and encounter lifecycles, persistent workflow state,

explicit audit records, and stable short-duration operation under 1-20 simulated devices. Functional validation produced no automated-test failure, false packet acceptance or rejection, database-integrity violation, or unexpected request error. Exploratory bench observations showed close displayed-value agreement with a consumer pulse oximeter, confidence-gated respiratory-rate availability, and low single-digit-watt input power for the tested node. Taken together, the results support EdgeER as a reproducible and technically credible multipatient reassessment engineering prototype. They do not establish diagnostic accuracy, medical-device performance, or suitability for patient-care decisions.

References

[1] Zacurate, "500 Series Fingertip Pulse Oximeter Blood Oxygen Saturation Monitor," consumer product page, https://www.amazon.com/dp/B07PQ8WTC4/ (accessed July 2026).

Appendix A. Configuration and Reproducibility

Configuration	Value
Offline timeout	15 s
Maximum packet age	300 s
Maximum future clock lead	60 s
Recheck interval	900 s
RR confidence gate	0.50
RR local demonstration range	8-30 breaths/min
Simulator scenarios	stable; yellow_low_oxygen; orange_trend; crashing; no_finger

Reproduction commands

```
python tools/software_validation/run_all_software_validation.py --quick
python tools/software_validation/run_all_software_validation.py --full --duration-seconds 60 --device-counts 1,5,10,20
python tools/software_validation/run_all_software_validation.py --full --duration-seconds 300 --device-counts 1,5,10,20
```

The validation package preserves raw CSV and JSON outputs, plots, screenshots, environment metadata, group summaries, and command-line logs under results/software_validation/.